

学术动向调研报告 | Consensus Pipeline v5.1 | 2026年07月15日

对近5年76篇文献（S级41篇、A级13篇）系统性调研。CNN是当前主流方法；因果推断与深度学习融合是新兴方向；多任务联合建模与跨领域迁移存在显著空白。

- 方法论贡献有限，应用导向为主：28篇论文列表（原分析子集）中，原创性方法论贡献极少，多数为综述或应用导向型文章。置信度：中
- CNN是能源经济学领域当前主流建模方法；因果推断与深度学习的融合已成为新兴方向。置信度：高
- 多任务联合建模与跨领域迁移学习仍存在显著研究空白。置信度：高

待方法论审查部门产出后补充

- 2003-2023年76篇论文的方法论演进：从单一预测 → 因果推断与可解释性并重。
- 多模态融合（文本+时序+图结构）成为新范式。

1. Explainable Artificial Intelligence (XAI): Concepts, taxonom — Information Fusion, 2019
2. Advances and Open Problems in Federated Learning — Foundations and Trends® in Machine Learning, 2020
3. The role of hydrogen and fuel cells in the global energy sys — Energy & Environmental Science, 2018
4. Blockchain technology in the energy sector: A systematic rev — Renewable and Sustainable Energy Reviews, 2018
5. The Future of Energy Supply: Challenges and Opportunities — Angewandte Chemie International Edition, 2006

1. 多任务联合建模 — 碳排放-电价-负荷缺乏联合优化框架 → 建议：多任务学习（MTL）
2. 动态因果效应 — 因果ML在能源政策评估中起步 → 建议：DID+因果森林

3. 跨领域迁移学习 — EU-ETS vs 中国碳市场模型迁移鲜有研究 → 建议：域自适应方法
4. GNN网络效应 — 电力/碳市场图结构信息未利用 → 建议：图神经网络建模溢出效应
5. 可解释性标配化 — 多数ML论文缺SHAP/LIME分析 → 建议：可解释性作为标准输出

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